Kernix 0.8.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

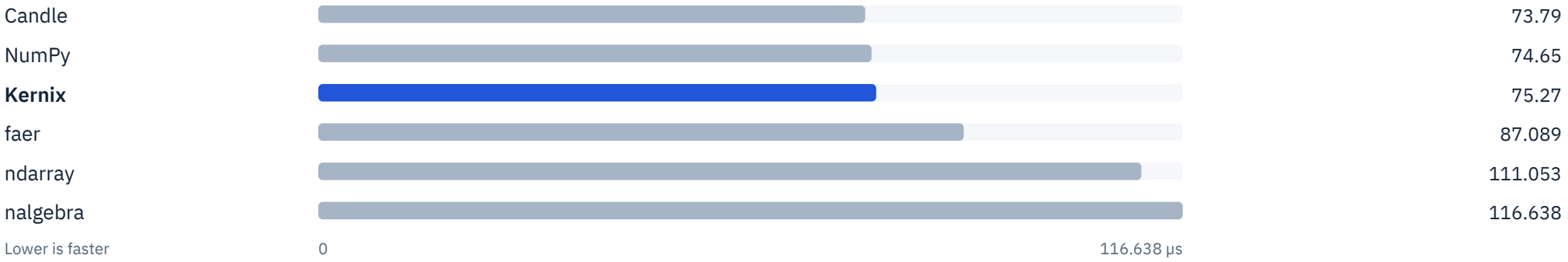
21 workloads

88 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

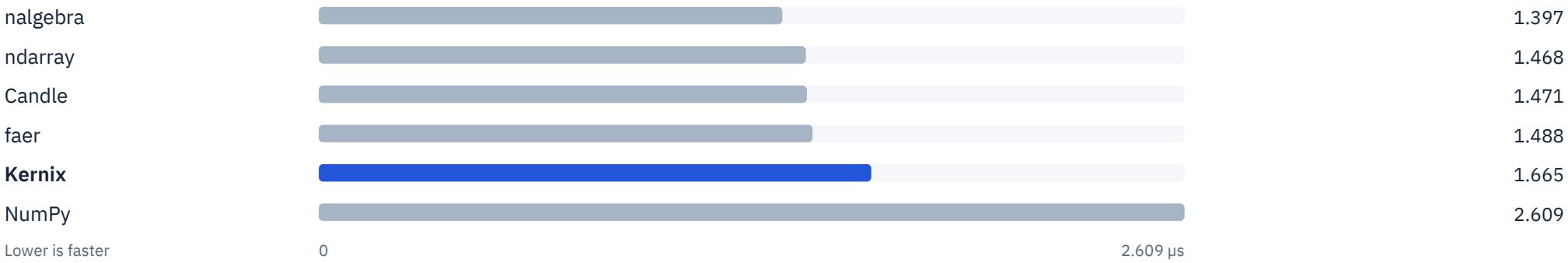
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

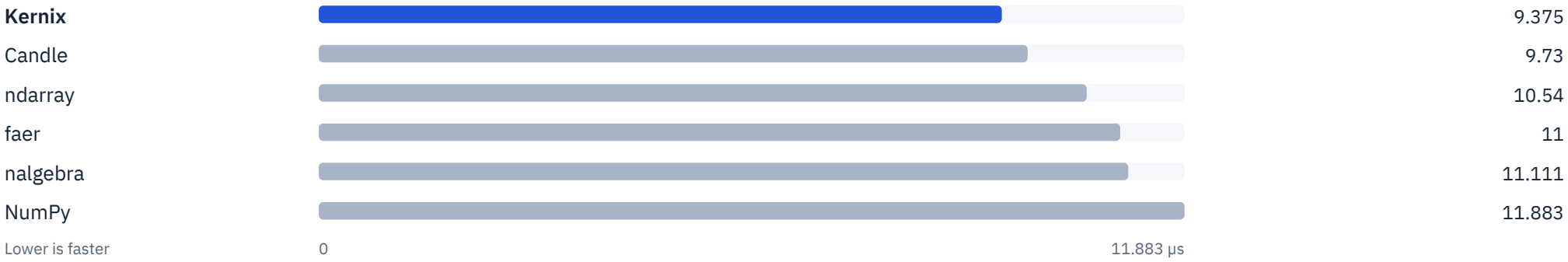
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



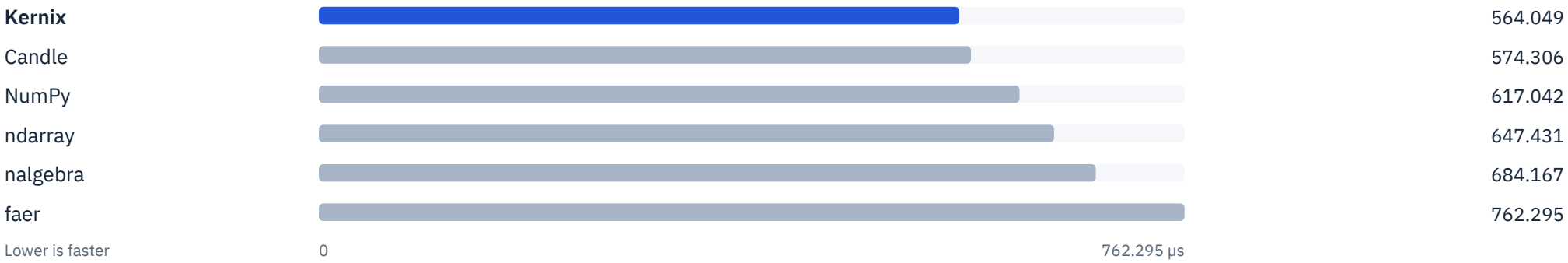
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

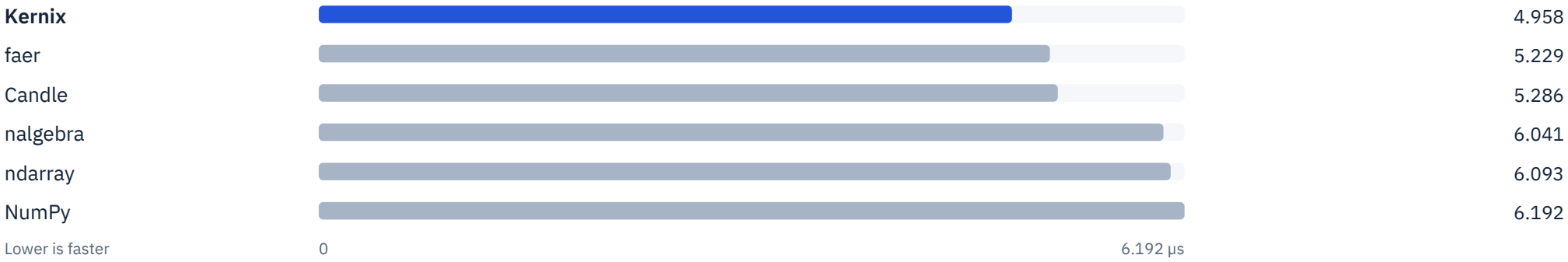
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



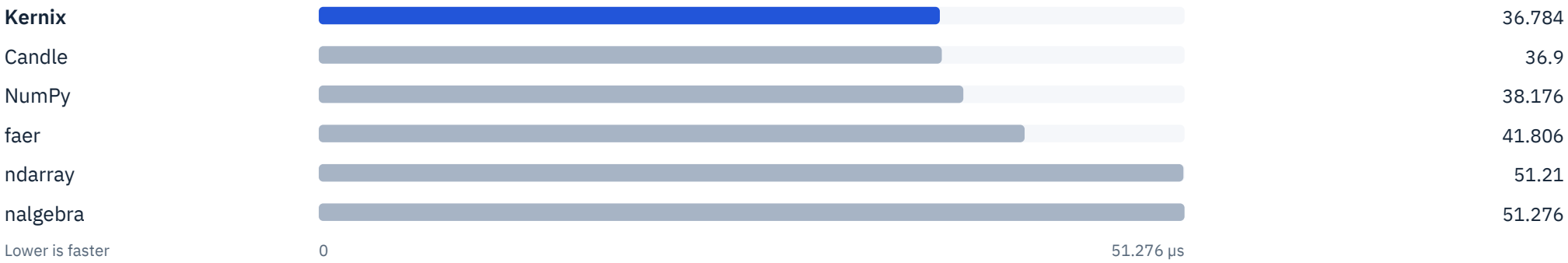
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

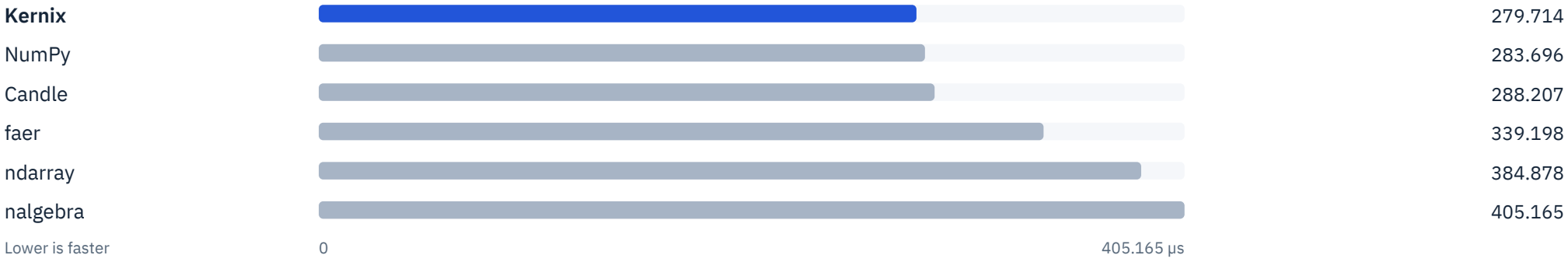
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

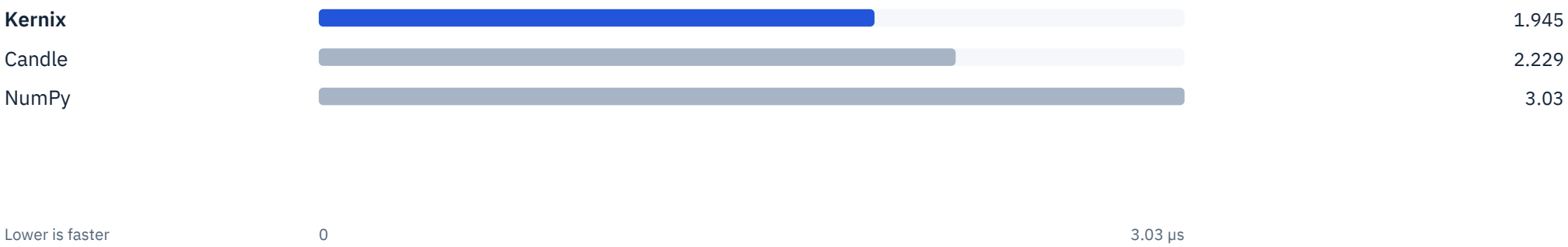


Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

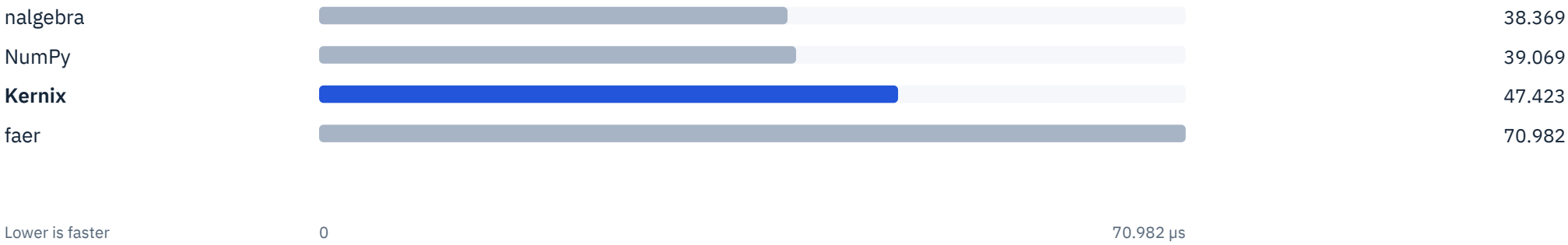


Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



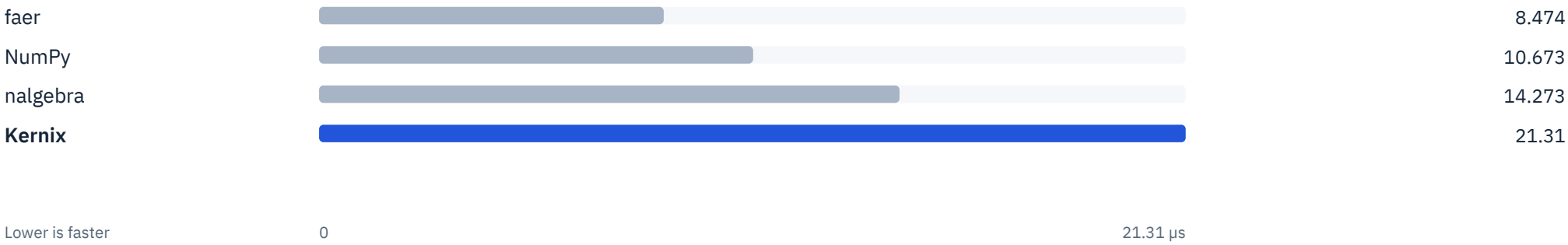
Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



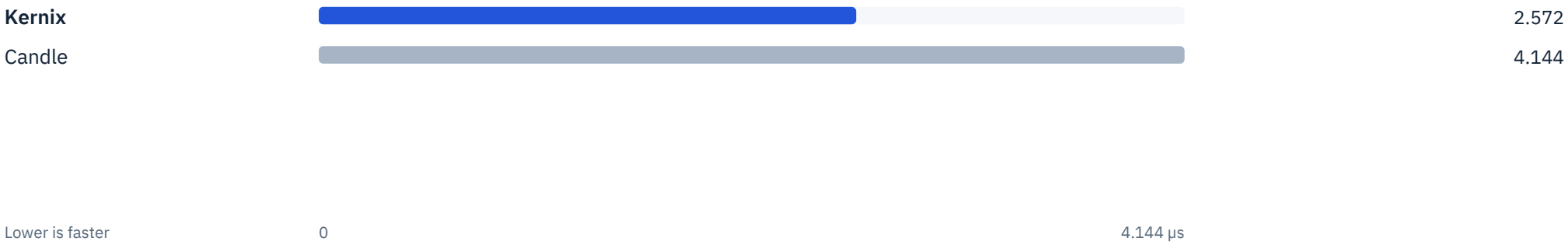
Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

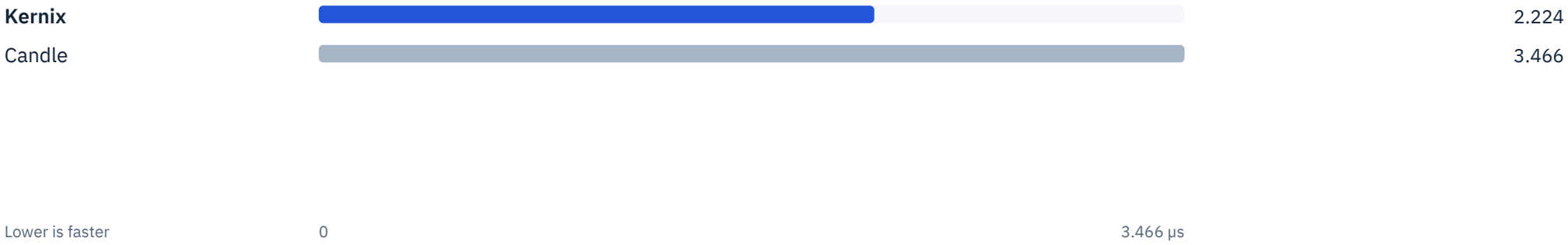
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

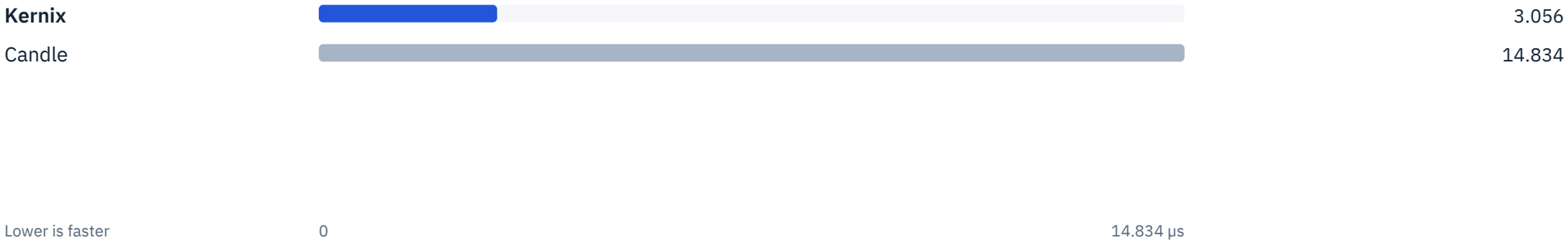
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

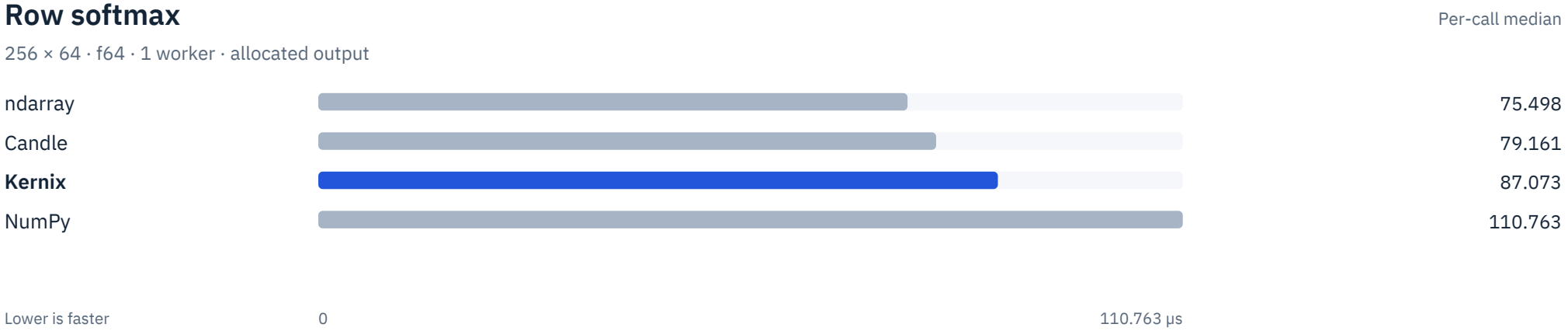


Kernix 0.8.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



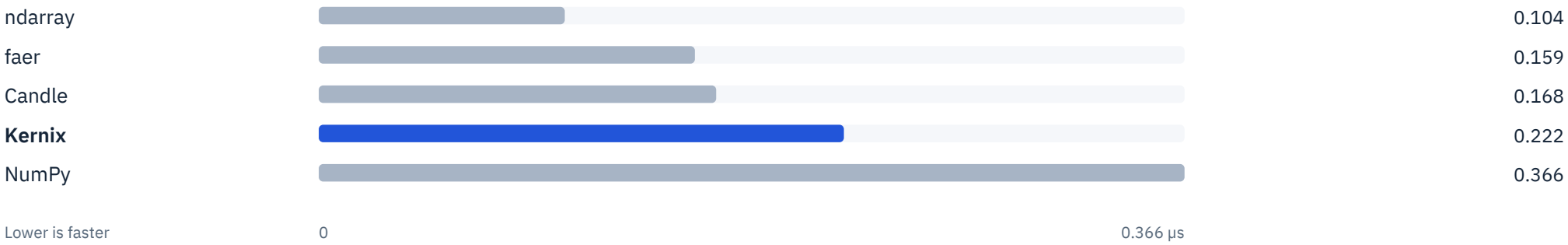
Kernix 0.8.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

64 × 64 · f64

Observed time reduction

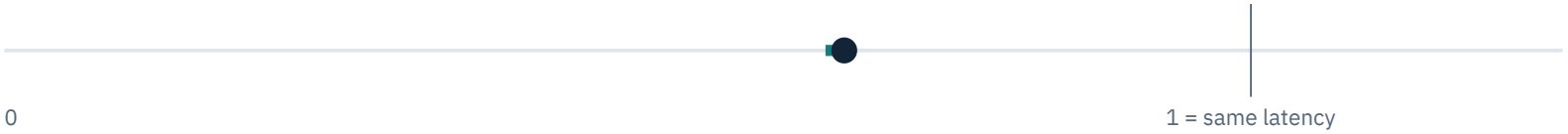


CURRENT / BASELINE	95% INTERVAL
0.8677118475888073	0.826547772693496 – 0.9410034528326016

Cholesky factorization

128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6737598467695953	0.6587561862739252 – 0.6826653795859267

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.49316287663357755	0.4813219013084302 – 0.5031802131103422

Thin QR

96 × 32 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0433176701999782	1.036087431693989 – 1.0842891742448475

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

256 × 64 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9581048304614215	0.9368562130504658 – 0.9876960858970454

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0148811150587593	1.0060131963811985 – 1.0317538528920642

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	**0.9939454550368949**	95% INTERVAL	0.9249689243870831 – 1.0027472527472527

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1665	1.665
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1468	1.468
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1488	1.488
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1471	1.471
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2609	2.609
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1397	1.397
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11111	11.111
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11883	11.883
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9375	9.375
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10540	10.54

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11000	11
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9730	9.73
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	116638	116.638
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	87089	87.089
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	111053	111.053
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	75270	75.27
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	74650	74.65
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	73790	73.79
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	684167	684.167
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	762295	762.295

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	617042	617.042
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	564049	564.049
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	574306	574.306
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	647431	647.431
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6041	6.041
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	5286	5.286
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4958	4.958
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6192	6.192
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5229	5.229
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6093	6.093

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51210	51.21
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	38176	38.176
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	51276	51.276
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	36784	36.784
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	36900	36.9
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	41806	41.806
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	339198	339.198
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	288207	288.207
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	405165	405.165
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	279714	279.714

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	283696	283.696
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	384878	384.878
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2229	2.229
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3030	3.03
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1945	1.945
Thin QR / Kernix 96 × 32 · f64	47423	47.423
Thin QR / faer 96 × 32 · f64	70982	70.982
Thin QR / nalgebra 96 × 32 · f64	38369	38.369
Thin QR / NumPy 96 × 32 · f64	39069	39.069
Thin QR / Kernix 256 × 64 · f64	484243	484.243

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	416558	416.558
Thin QR / nalgebra 256 × 64 · f64	372930	372.93
Thin QR / NumPy 256 × 64 · f64	348582	348.582
Cholesky factorization / nalgebra 64 × 64 · f64	14273	14.273
Cholesky factorization / Kernix 64 × 64 · f64	21310	21.31
Cholesky factorization / faer 64 × 64 · f64	8474	8.474
Cholesky factorization / NumPy 64 × 64 · f64	10673	10.673
Cholesky factorization / NumPy 128 × 128 · f64	72842	72.842
Cholesky factorization / nalgebra 128 × 128 · f64	98537	98.537
Cholesky factorization / faer 128 × 128 · f64	38466	38.466

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	121712	121.712
Cholesky factorization / faer 256 × 256 · f64	197958	197.958
Cholesky factorization / nalgebra 256 × 256 · f64	668195	668.195
Cholesky factorization / Kernix 256 × 256 · f64	633171	633.171
Cholesky factorization / NumPy 256 × 256 · f64	512867	512.867
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2572	2.572
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4144	4.144
Model inference / Candle batched normalized/sliced MLP · f64	3466	3.466
Model inference / Kernix batched normalized/sliced MLP · f64	2224	2.224
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2720	2.72

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	5085	5.085
Model backward / Candle prepared graph → fresh gradients · f64	14834	14.834
Model backward / Kernix prepared graph → fresh gradients · f64	3056	3.056
Parameter update / Kernix transactional SGD, fixed gradients · f64	529	0.529
Row softmax / NumPy 256 × 64 · f64	110763	110.763
Row softmax / Candle 256 × 64 · f64	79161	79.161
Row softmax / Kernix 256 × 64 · f64	87073	87.073
Row softmax / ndarray 256 × 64 · f64	75498	75.498
Create zeros / Kernix 64 × 64 · f32	222	0.222
Create zeros / faer 64 × 64 · f32	159	0.159

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	104	0.104
Create zeros / NumPy 64 × 64 · f32	366	0.366
Create zeros / Candle 64 × 64 · f32	168	0.168
Create ones / faer 64 × 64 · f32	203	0.203
Create ones / Kernix 64 × 64 · f32	204	0.204
Create ones / Candle 64 × 64 · f32	215	0.215
Create ones / NumPy 64 × 64 · f32	1349	1.349
Create ones / ndarray 64 × 64 · f32	141	0.141